# Data Science

## Introduction to APIs

GENERAL ASSEMBLY

# Lesson roadmap

- Welcome + Introduction
- APIs: The Basics
- HTTP: The Basics
- Submit HTML Requests
- Using A Python API
- Bring It Home

# Learning objectives

1. Explain what an application programing interface (API) is.
2. Explain the very basics of HTTP and how it's used to get data from the web.
3. Use the requests library to submit HTTP `requests` from Python.
4. Use a free Python API to access real-time stock prices.
5. Evaluate API documentation.

# APIs: The Basics

# What is an API?

API stands for **Application Programming Interface**.

In the broadest sense of the definition, an API is any pattern of programming that allows a programmer to solve a task.

There are other APIs that are commonly used by data scientists:

- The **pandas** API
- The **matplotlib** API

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# What is a Web API?

Ok, but this lesson is specifically about **Web APIs**. A web API uses (usually) **HTTP** to retrieve or manipulate data from the web.

**HTTP** = **H**yper**T**ext **T**ransfer **P**rotocol

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# Understanding the Web

When you're accessing a **website**, you are:

Sending a request to a server and getting back HTML, CSS and JavaScript files

When you're using a **web API**, you are:

Sending a request to a server and getting back structured data

This process is **calling an API** or **making an API request**.

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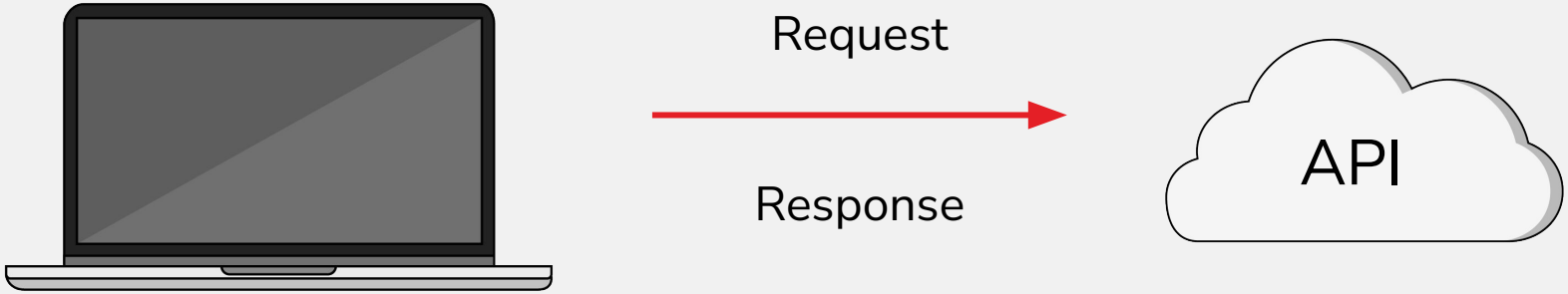
# APIs provide great data sources

**APIs** are data sources that can give you access to information from all sorts of places on the internet. Simply put, an API is a system that receives **requests** and returns **responses**.



# What Kind of Data Is Served Up With an API?

It'd be difficult to find data that **doesn't** have an API. Common uses are:

- **Social media integration**: When you post something on Instagram, you can automatically have it post to X, Snapchat, Facebook etc as well.
- **Payment APIs**: When you want to send or receive money via Venmo or the Cash App, the app directly interfaces with your bank or financial institution.
- **Maps**: Many apps like Uber, Lyft, and Waze are built on top of existing map services like Google Maps or Mapbox.

One extensive list of public APIs can be found here: https://github.com/public-apis/public-apis

And here are some API wrappers: https://github.com/realpython/list-of-python-api-wrappers

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# Why APIs?

| Data source | Spreadsheets | SQL databases | APIs | Webscraping |
|---|---|---|---|---|
| **Format** | XLS, CSV | Relational database | JSON | HTML |
| **How to access it?** | Excel | PostgreSQL, MySQL, SQLite | Web browser, Python, R | Import.io, PowerBI, Python, R, BeautifulSoup |
| **When would you use it?** | Someone sends you data in a spreadsheet.<br><br>A spreadsheet is available to download from a website. | Your company's customer data is stored in a SQL database. | High frequency data on (for example) share prices is available through an API.<br><br>A private or public organisation has made their data accessible via an API. | Data is embedded in a web page, or across multiple web pages. |
| **Advantages** | Familiar format.<br><br>Easy to open with Excel. | Can handle millions of rows of data.<br><br>Industry standard for databases. | Structured, clean data.<br><br>Data updates in real time.<br><br>Many organisations have APIs. | Saves lots of manual copy-pasting from websites into spreadsheet format. |
| **Disadvantages** | Data can be messy and unstructured.<br><br>Excel can only handle small files. | Data can't be analysed in depth using SQL. | Some APIs are not free to use.<br><br>Not all datasets are available through APIs. | Caveats around how to scrape ethically and responsibly.<br><br>Some websites are difficult to scrape. |

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# Requesting Data

# Calling an API: International Space Station data

Many APIs are easy to call. Just visit the URL in your browser!

Let's view how many people are in space right now.

In your browser, visit:

http://api.open-notify.org/astros.json

**Pro tip**: you can install this Chrome plug in for viewing the output nicely, or try this for Edge

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# JSON

The data returned by the API is in **JSON** (JavaScript Object Notation) format.

It looks a lot like a **Python dictionary**, and is treated in almost exactly the same way.

JSON consists of **key-value pairs** exactly like a dictionary, and often contains nested lists or dictionaries.

We already have all the knowledge we need to start making API requests in Python.


```
{
    "number": 10,
    "people": [
        {
            "name": "Sergey Prokopyev",
            "craft": "ISS"
        },
        {
            "name": "Dmitry Petelin",
            "craft": "ISS"
        },
        {
            "name": "Frank Rubio",
            "craft": "ISS"
        },
        {
            "name": "Stephen Bowen",
            "craft": "ISS"
        },
        {
            "name": "Warren Hoburg",
            "craft": "ISS"
        },
        {
            "name": "Sultan Alneyadi",
            "craft": "ISS"
        },
        {
            "name": "Andrey Fedyaev",
            "craft": "ISS"
        },
        {
            "name": "Jing Haiping",
            "craft": "Tiangong"
        },
        {
            "name": "Gui Haichow",
            "craft": "Tiangong"
        },
        {
            "name": "Zhu Yangzhu",
            "craft": "Tiangong"
        }
    ],
    "message": "success"
}
```

# APIs are just urls with parameters

API requests use HTTP, so the way to request information from an API may look somewhat familiar to you!

```
http://www.domain.com:1234/path/to/resource?a=b&x=y
```

It's a URL address, like what you type when you want to go to any website. But there's more here than meets the eye. Let's talk about each part of this address individually.

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# APIs are just urls with parameters

Port

Query

`http://www.domain.com:1234/path/to/resource?a=b&x=y`

Protocol

Host

Resource path

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# APIs are just urls with parameters

To allow us to extract subsets of data, APIs typically let us specify **parameters** (you can think of these as **options** or **filters**).

Port

Query

`http://www.domain.com:1234/path/to/resource?a=b&x=y`

Protocol

Host

Resource path

Parameter value

A ? separates the URL from the parameters

Parameter name/key

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# APIs usually offer a number of endpoints

These endpoints allow you to get the exact data you want in a useful structure, rather than having to get lots of extra data.

`http://www.domain.com/authors`

Endpoint

`http://www.domain.com/articles`

`http://www.domain.com/books`

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# When you interact with an API, you need to specify what type of action you want to take. These are HTTP Request Methods

## GET

- Retrieve information.
- Used by your browser's address bar every time you visit a website.

## POST

- Send information.
- Used by most forms to submit data, such as logging in or creating a new post.

## PUT

- Update information.
- Requests to change information in a database.

## DELETE

- Delete information.

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# What is HTTP?

HTTP is a set of rules for passing data around the web.

HyperText

**As in, HyperText Markup Language (HTML)**

Transfer

**Movin' stuff around**

Protocol

**A set of rules**

**Sidenote** You might have heard of a few **protocols** already:

- IP = Internet Protocol
- (S)FTP = (Secure) File Transfer Protocol (how files are sent from user to user)
- POP3 = Post Office Protocol 3 (how emails are sent)

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# The Example Example

Let’s check out this website:

www.example.com

And press **Ctrl-Shift-I** (**Cmd-Option-I** for Mac).

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# The Example Example

Now, let's open up our **command lines** and carry out our first HTTP request!

```
curl www.example.com
```

... whoa. Why did we get this?

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# HTTP Responses

HTTP responses are in the form of **just plain text**.

For most websites, this is HTML. If you want to **parse** this HTML, you could use an **HTML Parser** such as BeautifulSoup. (This is webscraping.)

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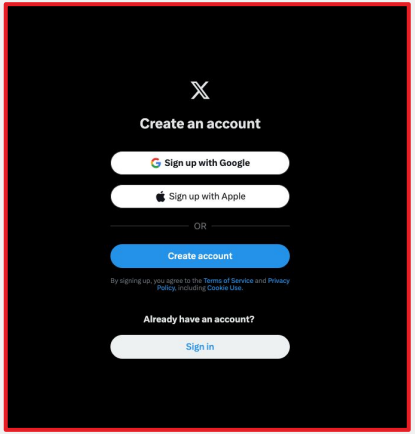
# What we do... literally every time we go to a website.

```
GET www.twitter.com
```

```html
<!doctype html>
<html>
<head>
   <title>Welcome to X or Whatever!</title>
</head>

<body>
    ...
</body>
</html>
```



# Requesting Data Using `requests`

Once you have identified an address and a method, you can send a request using Python's `requests` library. This will allow you to easily work with the data.

```
import requests

data = requests.get(address)
```

**What other methods might there be?**
**How would you find out?**

Requests library documentation

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# Informing a Marketing Strategy

You work for a competitor to **Ring**, the smart doorbell and security camera system.

To help inform their marketing strategy, we need to **find out about the prevalence of different types of crimes around the UK.**

We need to grab this data using an API.

- [ ] Open **02-APIs-marketing-strategy.ipynb**

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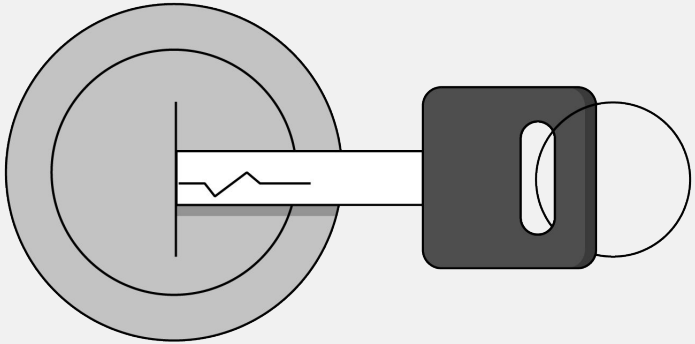
# Evaluating APIs

# API Keys

Many APIs require you to include a **key** with your request. You can think of this like a password that prevents someone from misusing the API.

Keys are also used to track your requests for rate limits and potential charges. Don't worry, most APIs have a free tier! You can register with the API's website to receive your unique key.



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# API Authentication: IMPORTANT

Most importantly, **never publish your API key!**

This includes on GitHub, **even if you're keeping your code private**.

There are other ways to provide your key to a server in order to keep that key safe, for example by storing it in a separate file, which is then not shared.

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# Read the Docs

Every API is unique — you will have to explore its **documentation** to figure out what data you want to access and what URL to use in order to send requests. However, there are a few standard details to check for in any API:

1. **Do I need an API key?**
    a. If so, are there rate limits or charges for requests?
    b. How do I include the key in my requests?
2. **What are the available endpoints for data?**
    a. How can I build a URL to get the exact data we want?
3. **What will the response data look like?**
    a. Typically, the response will be JSON.

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# What are the Risks?

Deciding to use an API requires careful consideration, especially if it's needed for a long term project or a project that needs to be scalable. Bear in mind:

1. **Costs**
   If your API usage is currently free, will you hit the free tier limit and need to pay in future? Is this affordable as your usage scales up?

2. **Maintenance**
   Will this API still be available in the future? How would your work be affected if it became unavailable or stopped being maintained?

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# Which Organisations Have APIs?

There's no comprehensive directory of every API in the world. The process of checking if there's an API to meet your needs will involve a bit of clever Googling.

Every API should have **documentation** which shows us how to construct an API request (i.e. a URL) that will retrieve exactly the data we want.

In breakout rooms, take a look at these APIs and find out what sorts of data or services they provide. **Are they free to use? Why would the organisation be motivated to make their data available through APIs?**

1. AlphaVantage
2. Transport for London
3. Google Cloud Vision
4. Google Cloud Language

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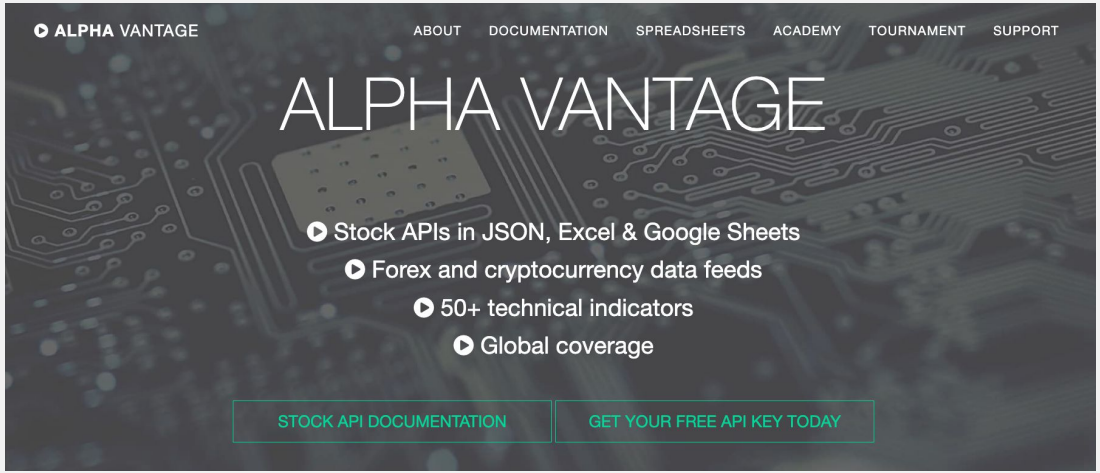
# Share price API

# Alphavantage

This is an API that provides live, free (up to a limit) data on share prices, forex, and cryptocurrencies. Let's explore the documentation:
https://www.alphavantage.co/

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# Share price API

Let's try out a demo API request from the documentation:

https://www.alphavantage.co/query?function=TIME_SERIES_INTRADAY&symbol=MSFT&interval=10min&apikey=demo

Looking at the **parameters** in the URL, what information do you think we're requesting?

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# Share Price API

Let's walk through the AlphaVantage API by doing 3 exercises:

Looking at share prices, looking at Forex, and looking at share prices in your portfolio.

- ☐ Open **03-APIs-financial-markets.ipynb**

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# Bring It Home

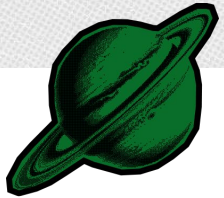
# Key takeaways

**APIs are very useful**

A safe, legal way of accessing large amounts of structured data

**You can access Web APIs easily**

Using the `requests` library and choosing your HTTP Method(s)



**There are a couple of key considerations when using them**

Never share API Keys with anyone, and consider costs and maintenance

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